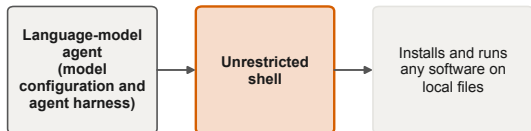


a

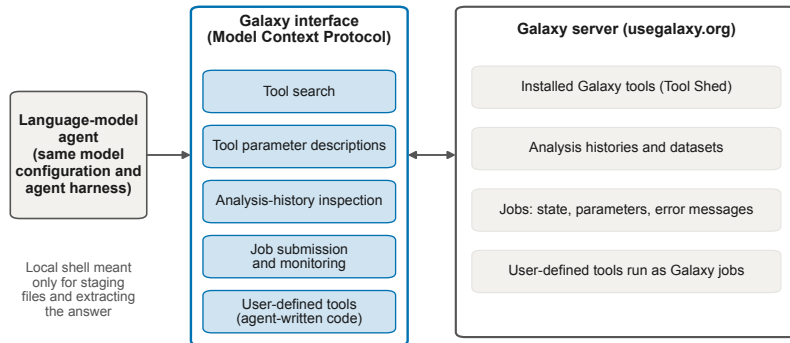
Open-ended code condition (reference)



Execution trace: commands, scripts, command output and final files; no record of individual analysis jobs

b

Galaxy condition (Galaxy-mediated execution)



Local shell meant only for staging files and extracting the answer

Execution trace and analysis history: tool identities and versions, requested and resolved parameters, datasets, job states and error messages

c Three benchmarks, five model configurations, three replicate runs of every task

Benchmark	BixBench-Verified-50		CompBioBench		IWC	
	Open-ended code	Galaxy	Open-ended code	Galaxy	Open-ended code	Galaxy
Execution condition						
GPT-5.5	150	150	300	300	30	30
GPT-5.6 Sol	150	150	300	300	30	30
GPT-5.6 Luna	150	150	300	300	30	30
DeepSeek V4 Pro (Codex)	150	150	300	300	30	30
DeepSeek V4 Pro (Claude Code, superseded; reported separately)	150	150	—	—	—	—
GPT-6 Astra (open-ended code condition only; not paired)	—	—	100	—	—	—

Numbers are runs: tasks × 3 replicate runs, per model configuration and execution condition

BixBench-Verified-50

50 tasks; platform-neutral benchmark
Endpoint: accuracy (share of runs scored correct by the benchmark evaluator)

CompBioBench

100 tasks; platform-neutral benchmark
Endpoint: reported benchmark score (answers credited, of 100)

IWC (Intergalactic Workflow Commission)

10 tasks; workflow-derived benchmark
Endpoint: output agreement with the workflow's reference output (0 to 1)

Archive: 4,240 runs · 4,228 execution traces
23,080 Galaxy analysis jobs · 69,812 Galaxy interface calls